

RECON ROVER V2 - ENTERPRISE INSTALLATION & OPERATION MANUAL

Version: 1.0.0 Architecture Version: 8.9 Document Version: 1.0 Project Repository:
Recon Rover V2

Revision History

Date	Version	Description	Author
2026-07-16	1.0	Initial Release of Enterprise Installation Manual	AI Engineering Team

1. Executive Summary

This document is the definitive build and operation manual for the Recon Rover V2. Derived exclusively from the Phase 8.9 repository implementation, it details the precise physical construction, wiring, software provisioning, and AI dependency installation required to replicate the autonomous platform.

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3. Hardware Requirements

- Microcomputer: Raspberry Pi 5 (8GB)
- Microcontroller: ESP32 DevKit V1

- Motor Controller: L298N Dual H-Bridge
- Sensors: HC-SR04 (Ultrasonic), MPU6050 (IMU), Pi Camera Module 3
- Power: 3x 18650 Li-Ion (3S BMS), 2x LM2596 Buck Converters

4. Software Requirements

- OS: Raspberry Pi OS (64-bit, Bookworm)
- Python: 3.12 +
- AI: Ollama, PyTorch, ONNX Runtime
- ESP32: PlatformIO Core

5. GPIO Tables

Raspberry Pi 5

Pin	Direction	Voltage	Connected Device	Purpose
GPIO 14 (TX)	OUT	3.3V	ESP32 RX (16)	UART Comms
GPIO 15 (RX)	IN	3.3V	ESP32 TX (17)	UART Comms
GPIO 2 (SDA)	I/O	3.3V	MPU6050	I2C Data
GPIO 3 (SCL)	OUT	3.3V	MPU6050	I2C Clock

ESP32 DevKit V1

Pin	Direction	Voltage	Connected Device	Purpose
GPIO 16 (RX2)	IN	3.3V	Pi TX	UART Comms
GPIO 17 (TX2)	OUT	3.3V	Pi RX	UART Comms
GPIO 25	OUT	3.3V	L298N ENA	Left Motor PWM
GPIO 26	OUT	3.3V	L298N ENB	Right Motor PWM
GPIO 27	OUT	3.3V	L298N IN1	Left Motor FWD
GPIO 14	OUT	3.3V	L298N IN2	Left Motor REV
GPIO 12	OUT	3.3V	L298N IN3	Right Motor FWD
GPIO 13	OUT	3.3V	L298N IN4	Right Motor REV
GPIO 32	OUT	3.3V	HC-SR04 Trig	Ultrasonic Ping
GPIO 33	IN	3.3V	HC-SR04 Echo	Ultrasonic Recv

(Note: ESP32 requires a logic level converter or voltage divider on the Echo pin if powered by 5V).

6. Wiring Guide & Power Architecture

1. **Battery Pack:** 3S (11.1V - 12.6V) connects to Dual LM2596 inputs.
2. **Buck 1 (5V):** Powers Raspberry Pi 5 (via USB-C or GPIO 5V) and HC-SR04 VCC.
3. **Buck 2 (5V):** Powers ESP32 (via VIN) and MPU6050 VCC.
4. **Direct (12V):** Powers L298N 12V input terminal (with onboard 5V jumper removed if > 12V, but 3S is safe).

5. Grounding: ALL grounds (Pi, ESP32, L298N, Sensors, Bucks, Battery) MUST be tied together.

7. Raspberry Pi Setup

Commands:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3-pip python3-venv git i2c-tools
```

Verification: Run `i2cdetect -y 1`. Expected output shows 0x68 (MPU6050).

8. Repository Setup

```
git clone <repository_url> ReconRoverV2
cd ReconRoverV2
python3 -m venv rover_env
source rover_env/bin/activate
pip install -r requirements.txt
```

9. AI Dependency Installation

```
curl -fsSL https://ollama.com/install.sh | sh
ollama serve &
ollama pull llama3:8b
```

(Verification: `ollama run llama3:8b "test"` responds successfully).

10. Firmware Upload (ESP32)

Commands:

```
cd "MAIN CODE/ESP32"
pio run --target upload
pio device monitor -b 115200
```

(Verification: Terminal displays "Rover Firmware Booted. Awaiting Serial.").

11. Calibration

- 1. IMU Calibration:** Run `python TOOLS/calibration_launcher.py --module imu`. Keep the rover perfectly still.
- 2. Motor Tuning:** Elevate the chassis. Run `python TOOLS/calibration_launcher.py --module motors`. Verify both treads spin forward at 50% PWM.

12. Testing

Execute the overarching demo verification test:

```
cd "MAIN CODE/RASPBERRY_PI"  
python main.py --demo
```

(Verification: EventBus outputs "SystemReady" followed by simulated mission steps).

13. Operation

Startup: Power on 5V bucks. Wait for Pi to boot. Run `main.py`. **Shutdown:** Send `SIGINT` (Ctrl+C). The `DemoRuntime` gracefully halts motors and closes UART.

14. Troubleshooting

- **No Serial Comms:** Verify RX/TX are crossed (Pi TX -> ESP32 RX). Ensure grounds are tied.
- **LLM Timeout:** Verify `ollama serve` is running. Check thermal throttling via `vcgencmd measure_temp`.
- **Vision Crashes:** Ensure camera ribbon cable is seated. Run `libcamera-hello` to test underlying driver.

15. Safety & Maintenance

- **Safety:** Do not place fingers near motor gears during `DemoRuntime` operation.
- **Maintenance:** Periodically check 3S battery cell balance. Do not discharge below 9.6V.

(End of Installation Manual)